24V_Power_IN

J1 EH_B2B-EH-A
F1 0679H5000-01
+24V
D2 B360B-13-F
C1 330uF/50V
R1 100
J5 EH_B4B-EH-A
Switch
SW_LED
GND

Power Distribution

+24V
L1 22uH
+E5V
U6 ISL85410FRZ
3-40V
VIN, EN, SYNC, BOOT, SS, PHASE, COMP, PG, FB, FS, VCC, EXP, PGND
R20 75k
C5 4.7F/50V
C6 100nF/50V
C7 100nF/50V
R23 75k
C9 27pF/10V
R19 75k
5.1V
R21 0
R22 N.C
R15 10k
C8 1uF/10V
GND

Connectors

U1A NUCLEO-F411RE
+3V3, +E5V, +12V
RESET_S1, BOOT0, PA0, PA1, PA4, PA13, PA14, PA15, PB0, PB7, PC0/PB8, PC1/PB9, PC2, PC3, PC11, PC10, PC12, PC13, PC14, PC15, PH0, PH1, +3V3_S1, +5V_S1, E5V, VBAT, VDD, VIN_S1, IOREF_S1, PD2, GND_S1
cRPM_METER, PWM_SPEED, MISO, MOSI, READY_IN, DIR, DRV_EN, SW_LED, EMO, Lid_Sensor

U1B NUCLEO-F411RE
Phase_MON, TEMP, WP, SCL, SDA, ADC_CS, SCK, ADC_UPDATE, ENC_CS, pBuzzer
PA2, PA3, PA5, PA6, PA7, PA8, PA9, PA10, PA11, PA12, PB1, PB2, PB3, PB4, PB5, PB6, PB8, PB9, PB10, PB12, PB13, PB14, PB15, PC4, PC5, PC6, PC7, PC8, PC9, AVDD, U5V, AGND, GND_S2

I2C_EEPROM

Address – b1010000
+3V3
U4 24LC512T-I/SN
A0, A1, A2, GND, VCC, SDA, SCL, WP
R16 4.7k, R17 4.7k, R18 4.7k
GND

ESC POWER

J9 EH_B2B-EH-A
+24V
GND

ESC PWM

J8 EH_B2B-EH-A
PWM_SPEED
GND

Buzzer

+3V3
CMT-7525-80-SMT-TR LS1
D1 B360B-13-F
Q1 2N7002
R12 100
pBuzzer
GND

Lid Sensor

J2 EH_B3B-EH-A
+3V3
R13 10k
C4 100nF/50V
Lid_Sensor
GND

EMO_SW

+3V3
J4 EH_B2B-EH-A
R14 10k
EMO
GND

ENCODER_SPI

J7 EH_B6B-EH-A
+3V3
ENC_CS, SCK, MOSI, MISO
GND

12-bit 2ch DAC

+E5V
C3 100nF/50V
U2 MCP4922-E/SL
SDI, SCK, CS, LDAC, SHDN, VDD, VREFA, VOUTA, VREFB, VOUTB, AVSS
MOSI, SCK, ADC_CS, ADC_UPDATE, aSPEED
GND

Thermistor (NTC)

+3V3
J6 EH_B2B-EH-A
R8 10k
TEMP
GND

Motor Driver

Phase_MON
R4 100
R2 75k
R3 10k
D3 MM3Z3V3T1G
ESC Driver hall counter
+24V
U3 367661
W1, W2, W3, VCC, GND, HALL_5V, H1, H2, H3, Speed_SET, CUR_LIM, GND, DIR, EN, DigIN2, DigIN1, Ready
J3 EH_B8B-EH-A
R10 4.7k
R6 10k
R9 100
aSPEED
DIR
R7 100
Enable = HIGH
Disable = LOW
DRV_EN
READY_IN
R5 330
Ready -> ON
Fault -> OFF
D4 150060BS55040
Q3 2N7002
Lid_Sensor
Magnet O -> HIGH
Magnet X -> LOW
Q2 2N7002
EMO
EMR Off -> LOW
EMR On -> HIGH
C2 100nF/50V
R11 330
cRPM_METER
ESCON Driver hall counter
Forced stop with Lid & E-STOP
GND

Revision History
- V 1.1
  - [-] Change capacitor layout size to large
  - [-] Change resistor layout size to large
  - [-] Add zener diode to decrease heat
  - [-] Change BJT layout pinmap
  - [-] Change PWM reading pin to ADC pin
- V 1.2
  - [ ] Change NPN to N-FET mosfet.
  - [ ] Zener Diode to DC/DC Converter
  - [ ] BEMF Block
- V 1.3
  - [-] Change R,C,D size to SMD
  - [-] Embedded DC/DC

Omniscience	
Sheet: /	
File: Spct.kicad_sch	
Title: Spin Coater	
Size: A3	Date: 2023-03-14
KiCad E.D.A. 8.0.2	Id: 1/1
	Rev: Rev 1.0